

Annual Donor Advised Fund Report Study

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What are DAFs?

- Donor advised funds (DAFs) are charitable giving accounts used by donor advisors
- Not all DAF sponsors have their financial data made public by the time the annual report is released
- The Johnson Center has been tasked with predicting the unreleased data and releasing an end of year report





What's new?

Our original project could be improved.

- Splitting data into training and test sets
- Not throwing out good data

So what did this change?

- How we group the data
- The error values got higher





Splitting up the Data

Training Data: Data that we know, roughly 75% of the total dataset.

Test Data: Data that we don't know. The training set predicts this data, and the predictions are compared to the actual values to determine accuracy.

Seed: Ensures that our data is split up in the same way every iteration.



Research Questions

- Is there a better way to make predictions for the Donor Advised Fund report?
- What is the optimal number of groupings for the chosen method?





Our Data and Variables

The Johnson Center is currently responsible for publishing the following data from charitable organizations at the end of each tax year:

- ❖ **Accounts:** quantity of DAFs currently held inside a charitable organization
- ❖ **Contributions:** total amount of money contributed into all DAF accounts under one organization
- ❖ **Grants:** amount of money distributed from DAF accounts
- ❖ **Assets:** balance of all DAF accounts under one organization
- ❖ **Type/Subtype:** classification of charitable organization (e.g. National Charity, University, Hospital, Nonprofit, etc.)

Data Cleaning

We were given two spreadsheets containing recent and historical data from the Johnson Center, which we merged using Python.

After reformatting and combining, we were left with ~23,000 rows of data, and while cleaning the data we removed:

- Imputed values
- Entries with N/A values
- Organizations with 0 accounts
- Negative monetary values

We eventually created a spreadsheet with ~9000 rows (about 35% of the data) and 8 columns



Results and Findings



Johnson Center Imputation Method

of DAF Accounts

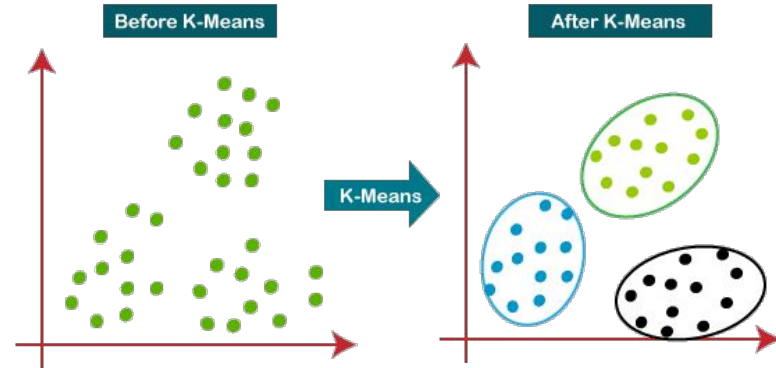
- AKA the weighted average growth method
- Group the data by year, type, and subtype
- Calculate the total growth rate for each group
- Apply each group's growth rate to each sponsor in that group

DAF Sponsor	Year 1	Year 2
Sponsor 1	400	500
Sponsor 2	25	25
Sponsor 3	25	25
Sponsor 4	25	25
Sponsor 5	25	25
Total	500	600

$$\text{Group Growth Rate} = \frac{600 - 500}{500} = .20$$

K-means Clustering Method

- For each metric, use the k-means algorithm to fit the training data into groups, and then predict where the test data falls into these groups
- Calculate the percent change for each sponsor in the training set
- Find the median percent change of groupings
- Apply to all missing data in the group



Results of Methods

To compute error, we took the difference between our predicted values and the actual values and divided by the actual values. We then took the **median** of all computed errors.

$$\frac{|\text{Actual} - \text{Predicted}|}{\text{Actual}}$$

Johnson Center Method:

Account Error:	9.60%
Contributions Error:	51.97%
Grants Error:	34.50%
Assets Error:	15.31%

K-means Clustering Method (k=4):

Account Error:	4.71%
Contributions Error:	50.27%
Grants Error:	30.39%
Assets Error:	11.65%

Results of Methods

To compute error, we took the difference between our predicted values and the actual values and divided by the actual values. We then took the mean of all computed errors.

$$\frac{|\text{Actual} - \text{Predicted}|}{\text{Actual}}$$

Johnson Center Method:

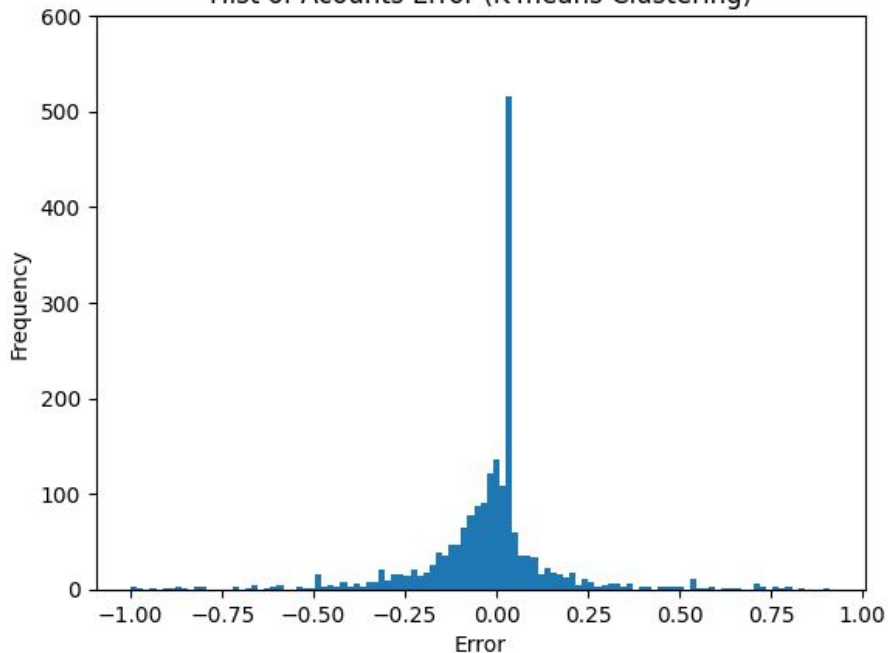
Account Error:	21.13%
Contributions Error:	328.61%
Grants Error:	187.52%
Assets Error:	96.06%

K-means Clustering Method (k=4):

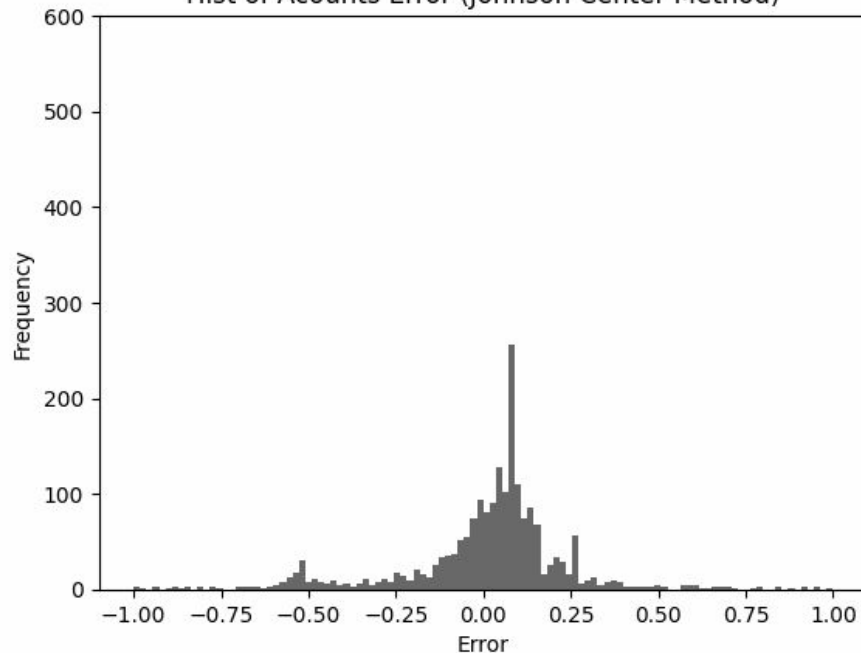
Account Error:	15.89%
Contributions Error:	277.48%
Grants Error:	159.37%
Assets Error:	91.63%

Quartile vs Weighted Average - Accounts

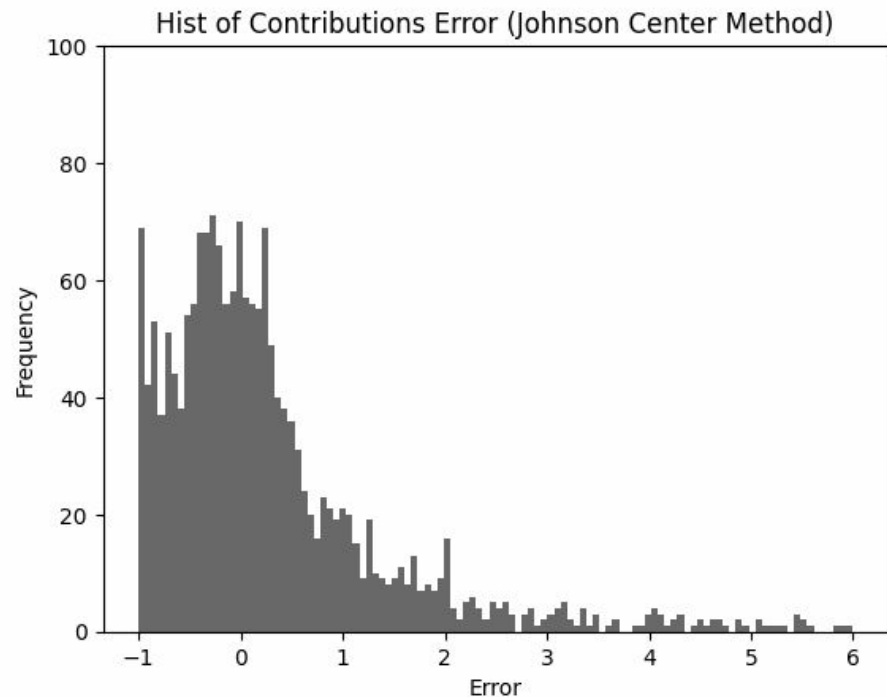
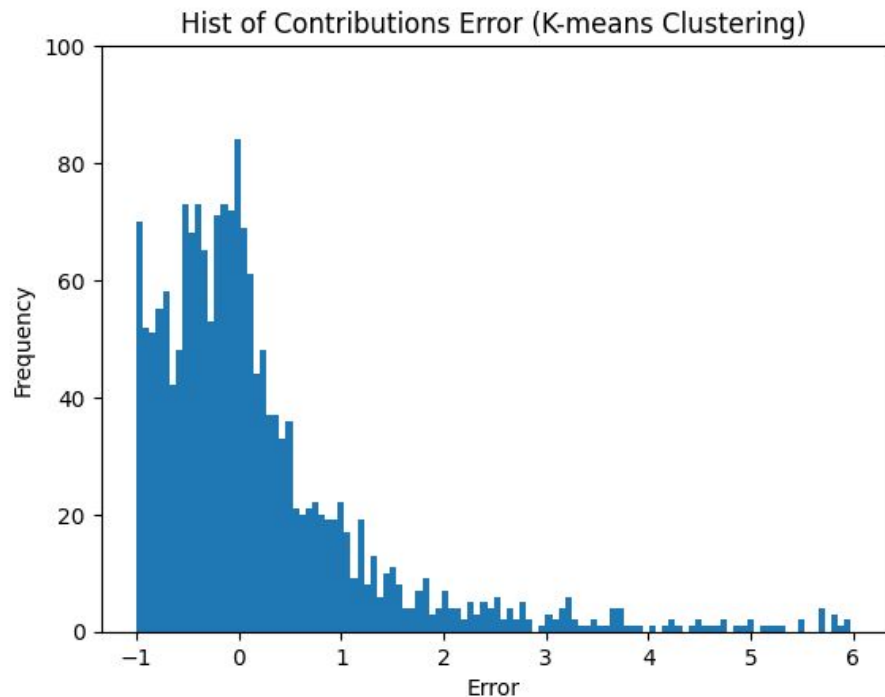
Hist of Accounts Error (K-means Clustering)



Hist of Accounts Error (Johnson Center Method)

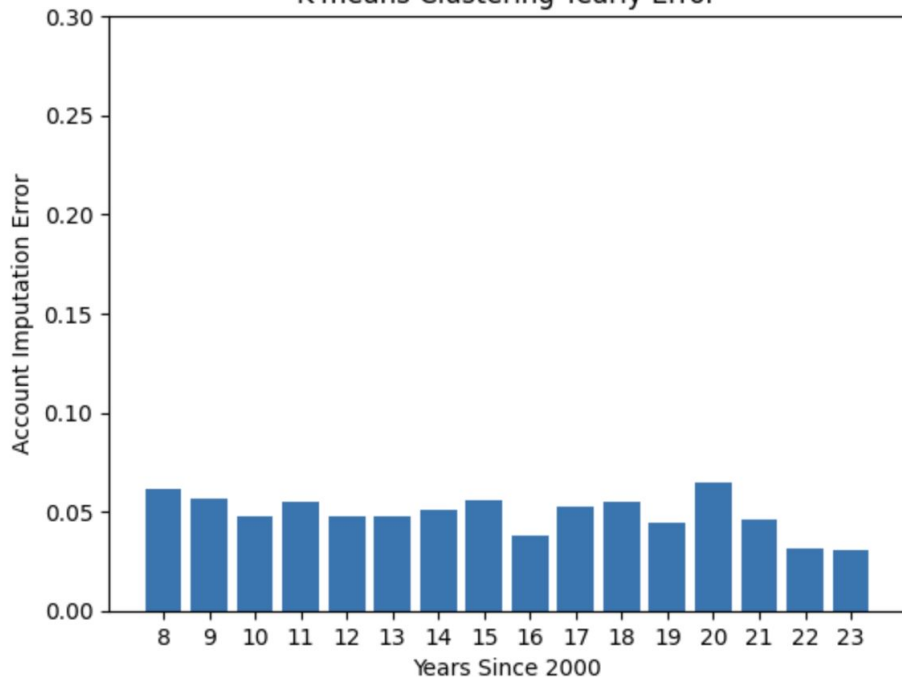


Quartile vs Weighted Average - Contributions

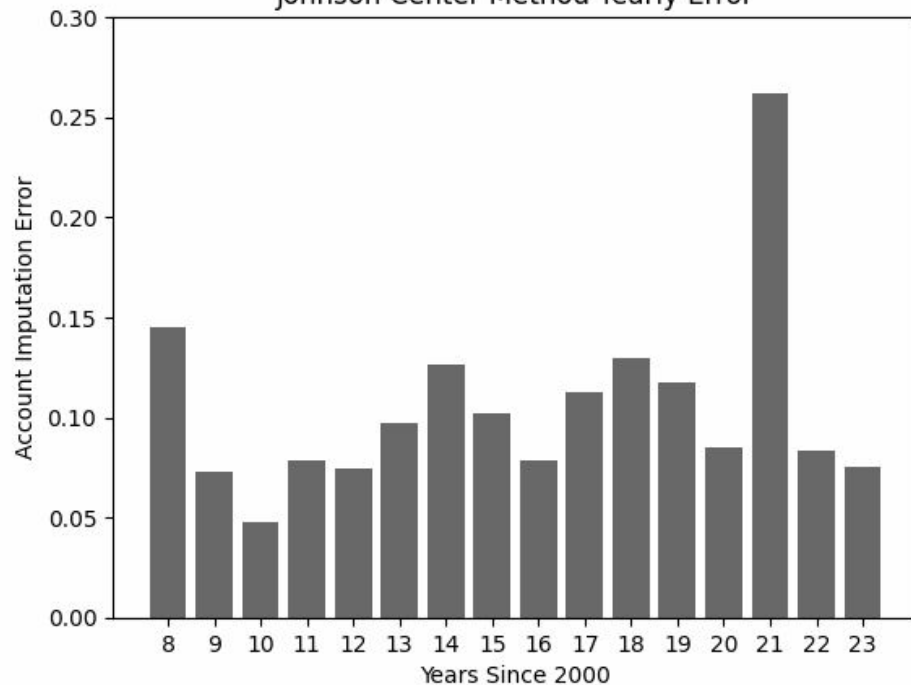


Yearly Error Comparison

K-means Clustering Yearly Error



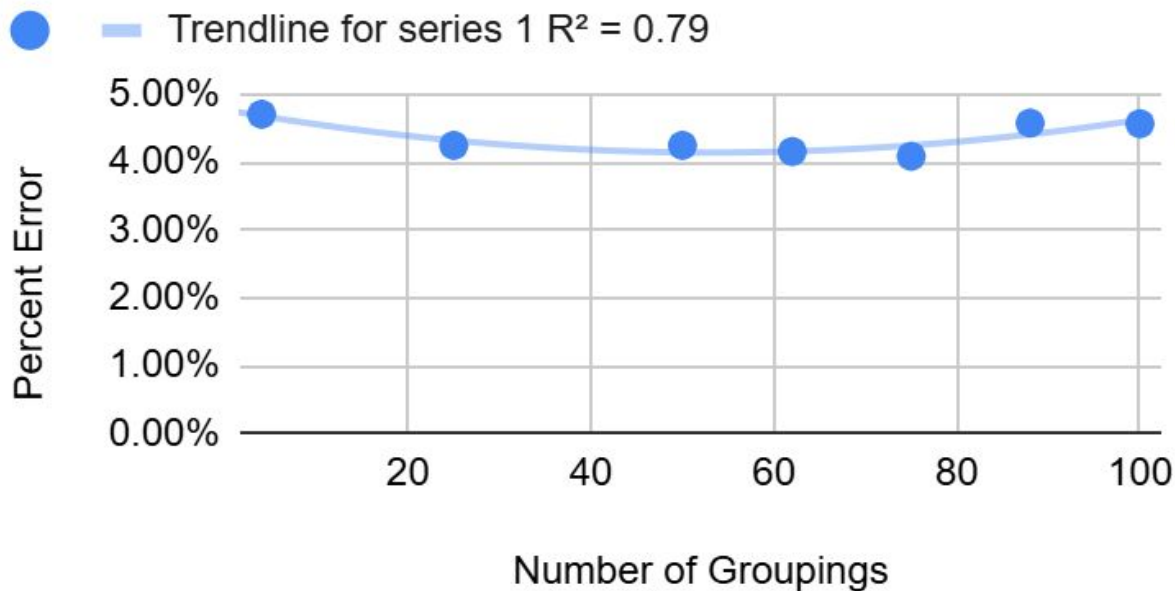
Johnson Center Method Yearly Error



Additional Testing

Number of groupings

K-Means Error for Full Dataset



accounts_group

0 8835

3 120

1 46

2 22

contributions

0 8941

2 72

1 8

3 2

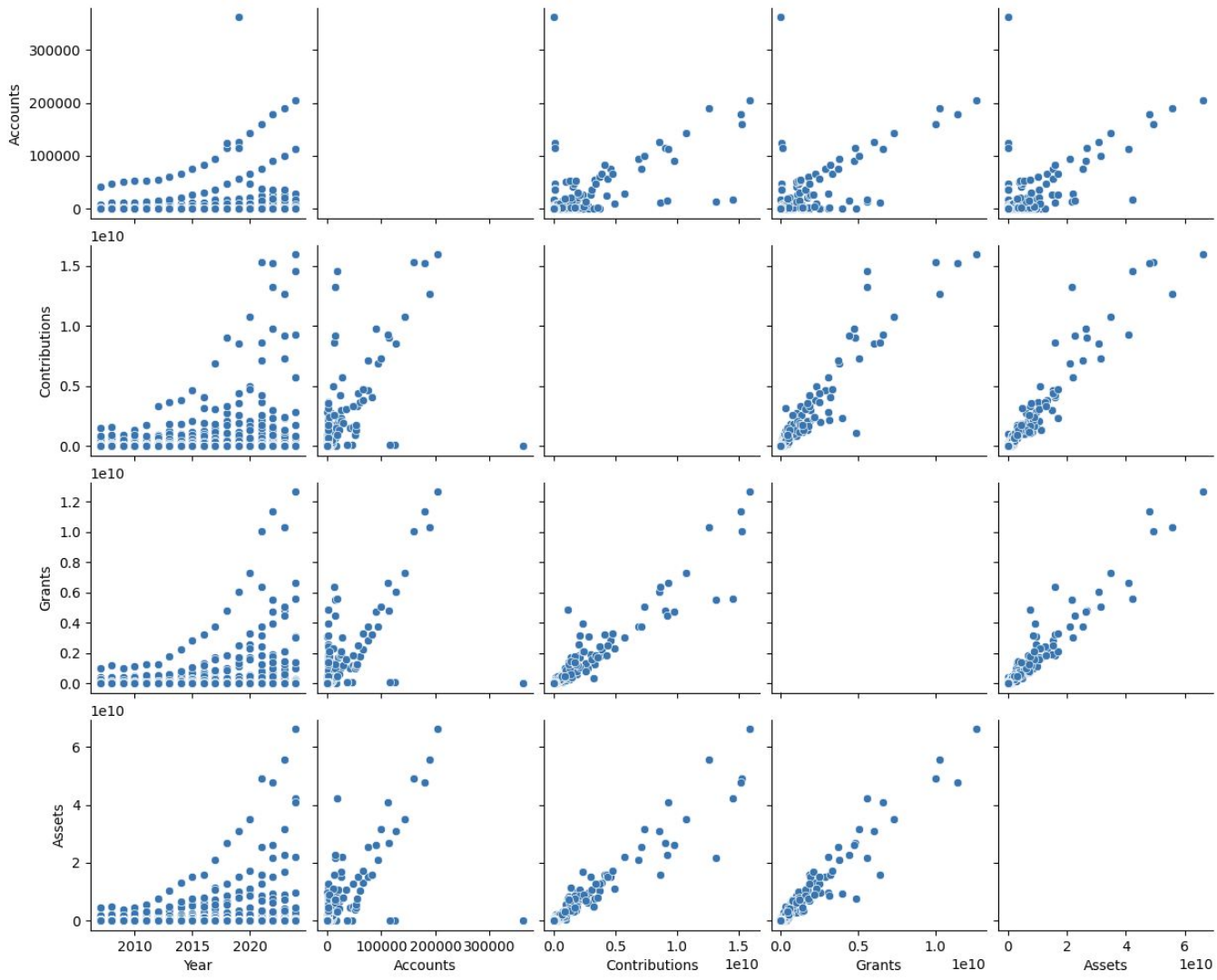
assets_group

0 8968

2 43

1 11

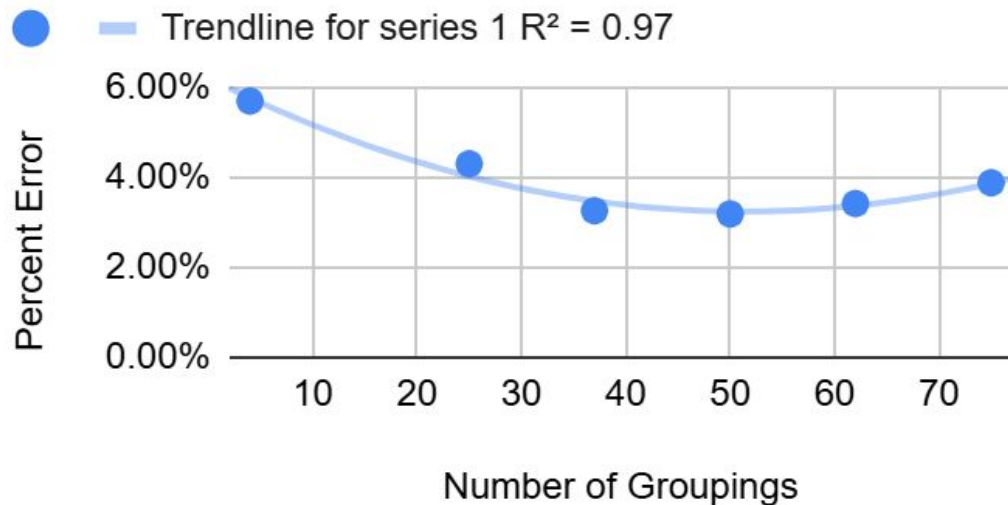
3 1



25 groups

Groupings for 2022-2023 Data

K-Means Error for 2022-2023 Data



assets_group

0 1074

22 142

15 56

17 32

9 21

11 8

16 7

24 7

23 6

6 5

21 4

18 3

7 3

1 3

3 2

10 2

14 2

20 1

5 1

2 1

12 1

4 1

13 1

8 1

19 1

K-Means Clustering vs. Quartiles

K-means clustering seems to slightly outperform static quartiles.

K-means Clustering (k=4):

Account Error:	4.71%
Contributions Error:	50.27%
Grants Error:	30.39%
Assets Error:	11.65%

Static Quartiles:

Account Error:	5.06%
Contributions Error:	53.57%
Grants Error:	32.55%
Assets Error:	12.08%

Additional k-means testing

- Grouping with scaled vs. unscaled data
 - Scaled performs slightly better
(although further research is needed)

Scaled K-means Clustering:

Account Error:	4.71%
Contributions Error:	50.27%
Grants Error:	30.39%
Assets Error:	11.65%

Unscaled K-means Clustering:

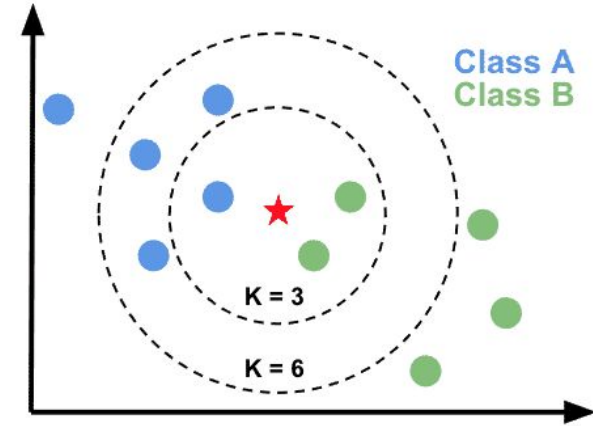
Account Error:	4.80%
Contributions Error:	50.38%
Grants Error:	30.39%
Assets Error:	11.79%

Other Attempted Methods



K-nearest neighbors

- Predicts the percent change of the test data by looking at the 'k' nearest training data points by metric, and finding the average percent change of those neighbors
- Got a median error that was a bit higher than for k-means, but the mean error was horribly high, so it was likely being strongly influenced by outliers



Conclusions





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**Thank you for
listening!**

